

Functions

Introduction to Programming

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Introduction

Functions

- **Function:** a name for some code stored in memory
- Function AKA method, procedure, subroutine
- Technically methods etc. differ from functions

Declaration

Functions

- Pseudocode, e.g.:
 - FUNCTION Greet()
 - OUTPUT "Hello!" to stdout
 - OUTPUT "How are you?" to stdout
 - ENDFUNCTION

Declaration

Functions

- PY: `def greet():`
- PY: `print("Hello!")`
- PY: `print("How are you?")`

Declaration

Functions

- JS: `function greet() {`
- JS: `console.log("Hello!");`
- JS: `console.log("How are you?");`
- JS: `}`

Declaration

Functions

- JV: `void greet() {`
- JV: `System.out.println("Hello!");`
- JV: `System.out.println("How are you?");`
- JV: `}`

Declaration

Functions

- C#: `void Greet() {`
- C#: `Console.WriteLine("Hello!");`
- C#: `Console.WriteLine("How are you?");`
- C#: `}`

Invocation/Call Functions

- Pseudocode, e.g.:
 - CALL Greet()

Invocation/Call Functions

- PY: greet()
- _ _ _ _ _
- JS: greet();
- _ _ _ _ _
- JV: greet();
- _ _ _ _ _
- C#: Greet();

Invocation (order of execution)

Functions

- `def greet():`
- 3. `print("Hello!")`
- 4. `print("How are you?")`
- 1. `print("Welcome to my program")`
- 2. `greet()`
- 5. `print("Goodbye")`

Parameters

Functions

- **Parameter:** a name for some data passed to a function
- A parameter is a variable
- A parameter is assigned a value when the function is called

Parameters

Functions

- Pseudocode, e.g.:
 - FUNCTION Greet(Name)
 - OUTPUT "Hello" + Name to stdout
 - OUTPUT "How are you?" to stdout
 - ENDFUNCTION

Parameters

Functions

- PY: `def greet(name):`
- PY: `print("Hello " + name)`
- PY: `print("How are you?")`

Parameters

Functions

- JS: `function greet(name) {`
- JS: `console.log("Hello " + name);`
- JS: `console.log("How are you?");`
- JS: `}`

Parameters

Functions

- JV: `void greet(String name) {`
- JV: `System.out.println("Hello " + name);`
- JV: `System.out.println("How are you?");`
- JV: `}`

Parameters

Functions

- C#: `void Greet(string name) {`
- C#: `Console.WriteLine("Hello " + name);`
- C#: `Console.WriteLine("How are you?");`
- C#: `}`

Arguments

Functions

- **Argument:** some data passed to a function at call time
- The argument is the value assigned to the parameter

Arguments (literals)

Functions

- PY: `greet("Harold")`
- `-----`
- JS: `greet("Harold");`
- `-----`
- JV: `greet("Harold");`
- `-----`
- C#: `Greet("Harold");`

Arguments (variables)

Functions

- PY: greet(harry)
- _ _ _ _ _
- JS: greet(harry);
- _ _ _ _ _
- JV: greet(harry);
- _ _ _ _ _
- C#: Greet(harry);

Arguments

Functions

- When declaring a function you choose the parameter names
- When calling a function the variables you pass to it as arguments need not have the same names as the corresponding parameters

Parameters (more than one)

Functions

- PY: `def greet(name, age):`
- PY: `yearOfBirth = 2024 - age`
- PY: `print("Hello " + name)`
- PY: `print("You were born in " + str(yearOfBirth))`

Arguments (more than one)

Functions

- PY: `greet("Harold", 75)`
- PY: `greet(name, age)`

Parameters (complex types)

Functions

- PY: `def print_all(array):`
- PY: `for element in array:`
- PY: `print(element)`

Parameters (complex types)

Functions

- JS: `function printAll(array) {`
- JS: `for (var element of array) {`
- JS: `console.log(element);`
- JS: `}`
- JS: `}`

Parameters (complex types)

Functions

- JV: `void printAll(Object[] array) {`
- JV: `for (Object element : array) {`
- JV: `System.out.println(element);`
- JV: `}`
- JV: `}`

Parameters (complex types)

Functions

- C#: `void PrintAll(object[] array) {`
- C#: `foreach (object element in array) {`
- C#: `Console.WriteLine(element);`
- C#: `}`
- C#: `}`

Return Value

Functions

- **Return value:** a value returned/output by a function
- A function can return only one thing
- The one thing returned may be complex, e.g. an array

Return Value

Functions

- Pseudocode, e.g.:
- FUNCTION GetSum(Numbers)
- SET Sum to 0
- FOREACH Number in Numbers
- COMPUTE Sum = Sum + Number
- ENDFOREACH
- RETURN Sum
- ENDFUNCTION

Return Value

Functions

- PY: `def get_sum(numbers):`
- PY: `sum = 0`
- PY: `for number in numbers:`
- PY: `sum += number`
- PY: `return sum`

Return Value

Functions

- JS: `function` getSum(numbers) {
- JS: `var` sum = 0;
- JS: `for` (`var` number `of` numbers) {
- JS: sum += number;
- JS: }
- JS: `return` sum;
- JS: }

Return Value

Functions

- JV: `int` getSum(`int`[] numbers) {
- JV: `int` sum = 0;
- JV: `for` (`int` number : numbers) {
- JV: sum += number;
- JV: }
- JV: `return` sum;
- JV: }

Return Value

Functions

- C#: `int GetSum(int[] numbers) {`
- C#: `int sum = 0;`
- C#: `foreach (int number in numbers) {`
- C#: `sum += number;`
- C#: `}`
- C#: `return sum;`
- C#: `}`

Return Value

Functions

- Pseudocode, e.g.:
 - SET Nums to [1, 2, 3, 4, 5]
 - COMPUTE Total = CALL GetSum(Nums)

Return Value

Functions

- PY: `sum = get_sum(numbers)`
- _____
- JS: `var sum = getSum(numbers);`
- _____
- JV: `int sum = getSum(numbers);`
- _____
- C#: `int sum = GetSum(numbers);`

Output as Input

Functions

- PY: `print(get_sum(numbers))`
- _____
- JS: `console.log(getSum(numbers));`
- _____
- JV: `System.out.println(getSum(numbers));`
- _____
- C#: `Console.WriteLine(GetSum(numbers));`

Variable Scope

Functions

- **Variable scope:** the durability and accessibility of a variable
- For how long does the variable exist in memory?
- To which other parts of the code (other functions) is the variable accessible?

Variable Scope

Functions

- **Global variable:** one that is declared outside of a function
- A global variables exists in memory for as long as the program runs
- A global variable is accessible to all program/module functions

Global Variables

Functions

- JS: `var global = 1;`
- JS: `console.log(global);`
- JS: `function printGlobal() {`
- JS: `console.log(global);`
- JS: `}`

Variable Scope

Functions

- **Local variable:** one that is declared inside a function
- A local variable exists in memory only while the function is executing
- A local variable is only accessible to its enclosing function

Local Variables

Functions

- JS: `function myFirstFunction() {`
- JS: `var first = 1;`
- JS: `}`
- JS: `function mySecondFunction() {`
- JS: `console.log(first); // runtime error`
- JS: `}`
- JS: `console.log(first); // runtime error`

Variable Scope

Functions

- Each global variable name must be unique
- Two or more local variables in the same function must have unique names
- A local variable may share its name with a local variable in another function

Modules and Libraries

Functions

- **Module:** a file containing related code
- Large programs are often comprised of many modules
- **Library:** a collection of modules

Procedural Programming

Functions

- A top-down approach to program design & development
- Break the problem down into smaller and smaller ones
- Keep going until the problem is small enough to solve with a function
- This process is known as functional decomposition